Infiltrative and Storage Disorders in the Cholestatic Infant: Hemophagocytic Lymphohistiocytosis, Niemann-Pick Disease Type C, Lysosomal Acid Lipase Deficiency, and Their Mimics.

Picture a six-week-old girl referred for jaundice. Her direct bilirubin, the conjugated fraction that the liver has processed but cannot excrete, is three milligrams per deciliter. Her stools are yellow. The resident who examines her writes, almost as an afterthought, that the spleen tip is palpable four centimeters below the left costal margin. That single sentence should change the whole work-up. A big liver is common in cholestasis. A big spleen in the first weeks of life is not, and it tells you that something is either filling the spleen or making it work too hard.

Now picture a ten-day-old boy, previously well, who arrives with fever, a falling platelet count and an alanine aminotransferase above three thousand units per liter. His ferritin, the iron storage protein that also rises with inflammation, comes back at twenty-five thousand micrograms per liter. And picture a seven-week-old who has been vomiting and has loose stools, who is not gaining weight, and whose abdominal ultrasound, ordered to look for pyloric stenosis, shows calcified adrenal glands.

The clinical question of this chapter is how to recognize, in an infant with cholestasis or liver failure, the diseases that infiltrate the liver from outside or fill it from inside. These are hemophagocytic lymphohistiocytosis, usually shortened to HLH, Niemann-Pick disease type C, lysosomal acid lipase deficiency, and a handful of conditions that look like them. The main idea is simple. Splenomegaly in a cholestatic infant means a storage or a blood disorder until proven otherwise. Each of these diseases has its own same-day test, most of them are diagnosed on blood rather than on a liver biopsy, and three of them, HLH, the infantile form of lysosomal acid lipase deficiency, and gestational alloimmune liver disease, are treatable only if you act within days.

A note on sources. For the cholestatic infant we use the 2017 joint guideline on cholestatic jaundice in infants, written by Fawaz and colleagues for the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, known as NASPGHAN, and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, known as ESPGHAN. For HLH we use the 2004 diagnostic criteria published by Henter and colleagues for the Histiocyte Society in 2007, the long-term results of the HLH-94 protocol reported by Trottestam and colleagues in 2011, the HLH-2004 results reported by Bergsten and colleagues in 2017, the emapalumab trial by Locatelli and colleagues in 2020, the ferritin study by Allen and colleagues in 2008, a Japanese survey of neonatal HLH by Suzuki and colleagues in 2009, a United States database study by Balakumar and colleagues in 2022, and the King's College Hospital series of infant liver failure by Hadzic and colleagues in 2022. For Niemann-Pick type C we use the 2018 international consensus by Geberhiwot and colleagues, a newborn series by Gumus and colleagues in 2017, and the drug trials by Mengel and colleagues in 2021 and Bremova-Ertl and colleagues in 2024. For lysosomal acid lipase deficiency we use the natural history study by Jones and colleagues, the enzyme replacement

trials reported by Jones in 2017 and Vijay in 2021, the ARISE trial by Burton and colleagues in 2015, and the dried blood spot assay described by Hamilton and colleagues in 2012. For gestational alloimmune liver disease we use the work of Rand, Whitington and colleagues.

Part one. One sign, many diseases.

The 2017 joint guideline puts it in one line: splenomegaly at two to four weeks of age should point toward storage or hematologic disorders. Think of four families. The first is HLH, either familial, caused by a genetic defect in how killer lymphocytes destroy their targets, or triggered, most importantly in the newborn by herpes simplex virus. The second is Niemann-Pick disease type C, a defect in moving cholesterol out of the lysosome, the cell's recycling compartment. The third is lysosomal acid lipase deficiency, which we will call LAL deficiency, whose infantile form is known as Wolman disease. The fourth is a group of mimics: gestational alloimmune liver disease, called GALD, which was once known as neonatal hemochromatosis; acid sphingomyelinase deficiency, the older Niemann-Pick types A and B; perinatal Gaucher disease; the transient abnormal myelopoiesis of Down syndrome; stage 4S neuroblastoma; Langerhans cell histiocytosis; and congenital leukemia.

These diseases share one bedside sign and almost nothing else, and each has its own first test. For HLH you send ferritin, triglycerides, fibrinogen, soluble interleukin-2 receptor, also called soluble CD25, and natural killer cell function and perforin expression. For herpes and enterovirus you send blood polymerase chain reaction tests. For Niemann-Pick type C you send a bile acid marker or a lipid marker called lyso-sphingomyelin 509, and then the genes. For LAL deficiency you measure enzyme activity on a dried blood spot. For acid sphingomyelinase deficiency you measure that enzyme. For GALD you look for iron outside the liver. For transient abnormal myelopoiesis you look at the blood film. For neuroblastoma you measure urine catecholamines, and for Langerhans cell histiocytosis you need tissue.

Part two. HLH: how a lymphocyte problem reaches the liver.

To understand HLH, start with how a normal cytotoxic lymphocyte works. Natural killer cells and cytotoxic CD8 T cells kill infected cells by releasing granules that contain perforin, a protein that punches holes in the target cell's membrane, and granzymes, enzymes that enter through those holes and trigger cell death. Killing does two things. It removes the infected cell, and it removes the antigen-presenting cells that keep the immune response switched on. In familial HLH, perforin itself is missing, or the machinery that moves the granules to the cell surface and fuses them with the membrane is broken. The genes, with their approximate share of familial cases, are PRF1, which encodes perforin, in about twenty to forty percent; UNC13D in about twenty to twenty-five percent; STX11 in about fourteen percent; and STXBP2 in about ten percent.

Here is the chain of events, step by step. The lymphocytes recognize the target but cannot kill it. The stimulus therefore never goes away. The lymphocytes keep releasing interferon gamma, which activates macrophages. Activated macrophages make ferritin, release cytokines, raise triglycerides by suppressing the enzyme that clears them, and consume fibrinogen. They also engulf red cells, white cells and platelets, which is the hemophagocytosis that gives the disease its name. Activated

lymphocytes and macrophages then infiltrate the organs, and the liver is a favorite target: they fill the sinusoids and the portal tracts, and the Kupffer cells, the liver's resident macrophages, become engorged. The result is hepatitis, cholestasis, coagulopathy and, in the worst cases, liver failure.

The liver is not a bystander in this disease. In the HLH-2004 study, aspartate aminotransferase was above one hundred twenty units per liter in seventy-six percent of patients and alanine aminotransferase in fifty-six percent at diagnosis. In the earlier HLH-94 study, jaundice, edema and a raised creatinine were associated with death before the child could reach stem cell transplantation.

Part three. Making the diagnosis of HLH.

The HLH-2004 criteria allow the diagnosis in one of two ways: a molecular diagnosis consistent with HLH, or at least five of eight clinical and laboratory criteria. The eight criteria are fever; splenomegaly; cytopenias affecting at least two of three lines, meaning hemoglobin below ninety grams per liter, an absolute neutrophil count below one thousand per microliter, or platelets below one hundred thousand per microliter; triglycerides of three millimoles per liter or more, or fibrinogen of one and a half grams per liter or less; hemophagocytosis in the marrow, spleen or lymph node without malignancy; low or absent natural killer cell activity; ferritin of five hundred micrograms per liter or more; and soluble CD25 of two thousand four hundred units per milliliter or more.

Each criterion needs a little translation for a jaundiced infant. A sick newborn may not mount a fever, and hypothermia should be read as a warning sign. Splenomegaly is the very sign that brings storage disease into the same differential. Neonatal hemoglobin norms are different from those of older children, and a falling platelet count is usually the earliest cytopenia. Low fibrinogen also occurs in any liver failure, so it must be read together with the other markers. Hemophagocytosis on a marrow sample is neither necessary nor sufficient. Natural killer cell testing takes days and should be sent together with perforin expression and a degranulation assay. Soluble CD25 is the most specific single marker of T cell activation.

How high a ferritin should worry you? Allen and colleagues at Texas Children's Hospital found that a ferritin above ten thousand micrograms per liter was ninety percent sensitive and ninety-six percent specific for HLH in their pediatric population. Remember, however, that ferritin is also very high in GALD and in other forms of neonatal liver failure, and that the practical rule is to start treatment on the pattern rather than waiting for the last confirmatory result.

Part four. HLH in the infant and the newborn.

How often does HLH hide behind liver failure in the youngest children? Hadzic and colleagues at King's College Hospital in London reviewed seventy-eight children under twenty-four months of age with acute liver failure. Thirty of them, thirty-eight percent, had the HLH phenotype. Nineteen of those thirty had a genetic finding, nine of them biallelic mutations in a primary HLH gene. Mortality among the children with genetic findings was about fifty-three percent, and about sixty-seven percent with biallelic mutations. In a United States inpatient database study of

seventy-six neonates with HLH, Balakumar and colleagues found liver dysfunction in thirty-five percent. Overall mortality was forty-two percent, but with liver dysfunction it rose to about sixty-eight percent, and liver dysfunction was the only independent predictor of death, with an odds ratio of six and a half. One in five of these newborns had hydrops.

The Japanese nationwide survey by Suzuki and colleagues described twenty neonates with HLH. Overall survival was forty percent. Six had familial HLH and one had HLH associated with severe combined immunodeficiency, and only two of those seven survived. Strikingly, six had HLH triggered by herpes simplex virus, as many as had familial disease. The practical lesson is to start acyclovir in any ill newborn with this picture before the polymerase chain reaction result is back. Herpes hepatitis is both an infection and an HLH trigger: treat the virus, measure soluble CD25 and fibrinogen, and add HLH therapy if the pattern persists.

How do you tell HLH from GALD, the commonest cause of liver failure at birth? In their review of neonatal acute liver failure, Taylor and Whitington estimated that GALD causes sixty to ninety percent of cases, viruses twenty to thirty percent, and HLH less than ten percent. The transaminases help. GALD is a disease of established, often prenatal, liver injury, so the infant is in liver failure from the first day with transaminases that are near normal and a high alpha-fetoprotein. Viral hepatitis and HLH, by contrast, are active, ongoing injuries with transaminases in the thousands. Ferritin is high in both and does not separate them.

Part five. Treating HLH.

The two international protocols tell a sobering story. In HLH-94, which used etoposide, dexamethasone and, later, cyclosporine, followed by stem cell transplantation, two hundred forty-nine patients had a five-year survival of fifty-four percent. Among those who were transplanted, five-year survival was sixty-six percent. Seventy-two patients, twenty-nine percent, died before they could be transplanted, and no child with familial HLH survived without transplantation. HLH-2004 moved cyclosporine to the start of therapy. In three hundred sixty-nine patients, five-year survival was sixty-one percent, compared with fifty-six percent in HLH-94, a difference that was not statistically significant. Death before transplantation fell from twenty-seven to nineteen percent, but that difference also missed significance. In verified familial HLH, survival was fifty-nine percent. The window between diagnosis and transplant is where children die, and liver failure is one of the reasons.

Emapalumab, a monoclonal antibody that neutralizes interferon gamma, was tested by Locatelli and colleagues in thirty-four patients, twenty-seven of whom had already failed conventional therapy. Sixty-three percent of the previously treated patients responded, well above the forty percent the investigators had set as the bar for success. It is a bridge, not a cure: familial HLH still needs a new immune system, which means stem cell transplantation, not a new liver.

Part six. Niemann-Pick disease type C.

Niemann-Pick disease type C is caused by mutations in NPC1 in at least ninety-five percent of cases and in NPC2 in the rest, with an incidence of about one in one hundred thousand births. To

understand it, follow a molecule of cholesterol. Low-density lipoprotein brings cholesterol into the cell and delivers it to the late endosome and lysosome. There it is freed from its ester. The NPC2 protein, which floats inside the lysosome, picks the free cholesterol up and hands it to NPC1, a protein in the lysosomal membrane, which moves it out to the rest of the cell. When either protein fails, unesterified cholesterol and glycosphingolipids accumulate inside the lysosome. Macrophages in the spleen, liver, marrow and lung fill up and become foam cells. In the developing brain the same trap slowly kills neurons.

In the newborn, the disease presents as prolonged cholestatic jaundice with an enlarged liver and spleen, sometimes after fetal ascites or hydrops discovered on a prenatal scan. The great trap is that the jaundice usually resolves by three to four months of age, while the spleen often stays large. According to the 2018 consensus, about eight to nine percent of children with this perinatal presentation progress to liver or multi-organ failure and die within six months, and across cohorts the fatal neonatal systemic form accounts for three to twelve percent of cases. The others seem to recover, and the diagnosis may not be made until the neurological disease appears at school age, with vertical gaze palsy, clumsiness, sudden loss of muscle tone with laughter called cataplexy, and cognitive decline. Statement nine of the consensus says that prolonged neonatal jaundice, hepatosplenomegaly or developmental delay under two years of age should raise the suspicion of Niemann-Pick type C.

When the disease begins at birth it is as much a lung disease as a liver disease. Gumus and colleagues in Ankara described ten newborn-onset cases, eight with NPC1 and two with NPC2 mutations. All ten had cholestasis and hepatosplenomegaly, nine had lung involvement, and two had fetal ascites. Nine of the ten died: six from pulmonary disease and three from liver failure. A cholestatic infant with rapid breathing, a big spleen and no heart disease needs Niemann-Pick type C on the list.

Diagnosis has changed. The first-line tests in the 2018 consensus are blood biomarkers. Oxysterols, oxidized forms of cholesterol, are raised, and the one called cholestane-triol is more specific than seven-ketocholesterol. But oxysterols are also raised in acid sphingomyelinase deficiency and LAL deficiency, less often in two other disorders of sterol metabolism, and, crucially for us, they can be spuriously raised in neonatal cholestasis itself. Lyso-sphingomyelin 509 is sensitive for both Niemann-Pick type C and acid sphingomyelinase deficiency, and a high ratio of lyso-sphingomyelin 509 to lyso-sphingomyelin favors type C. A bile acid, a trihydroxy-cholanoyl glycine, is raised only in Niemann-Pick type C and acid sphingomyelinase deficiency and is less affected by cholestasis, so it is preferred in the newborn, although it has not been validated prospectively. Filipin staining of cultured skin fibroblasts, once the gold standard, is no longer first line. In practice, for a jaundiced infant with a big spleen, send the bile acid marker or lyso-sphingomyelin 509 together with a cholestasis gene panel that includes NPC1, NPC2, the acid sphingomyelinase gene SMPD1 and the lysosomal acid lipase gene LIPA, and send the enzyme activities for acid sphingomyelinase and lysosomal acid lipase at the same time, because the biomarkers overlap.

Treatment has moved quickly, but only for the brain. Miglustat, which reduces the production of glycosphingolipids, is licensed in Europe, and the consensus advises against using it in patients without symptoms or with only liver and spleen enlargement. In a randomized trial of fifty patients reported by Mengel and colleagues, arimoclomol slowed progression on a five-domain severity scale by about sixty-five percent at twelve months, with a change of zero point seven six points compared with two point one five on placebo. The United States Food and Drug Administration approved it on the twentieth of September 2024 for use with miglustat from age two. In a sixty-patient crossover trial by Bremova-Ertl and colleagues, levacetylleucine improved an ataxia score by about one point three points more than placebo, and it was approved four days later, on the twenty-fourth of September 2024. None of these drugs has been shown to change the neonatal liver course. For the hepatologist, the value of making the diagnosis in infancy is genetic counseling, surveillance and early neurological treatment. In the rare infant with liver failure from Niemann-Pick type C, a new liver will not prevent the neurological disease, and the family needs to hear that before transplant is discussed.

Part seven. Lysosomal acid lipase deficiency.

Lysosomal acid lipase, encoded by LIPA, is the enzyme that breaks down cholesteryl esters and triglycerides delivered to the lysosome by lipoproteins. Without it, these lipids accumulate in hepatocytes, Kupffer cells, the intestinal wall and the adrenal glands. There is a second consequence. Because no free cholesterol leaves the lysosome, the cell senses that it is short of cholesterol, so it makes more cholesterol and more low-density lipoprotein. This is why the later-onset form of the disease looks like a lipid disorder, with high low-density lipoprotein cholesterol, low high-density lipoprotein cholesterol, and a fatty, enlarged liver that is easily mistaken for ordinary fatty liver disease.

The infantile form, Wolman disease, is devastating. In the natural history study by Jones and colleagues, thirty-five untreated infants had a median age at death of three point seven months, and only about eleven percent survived beyond twelve months. Adrenal calcification was seen in seventy-nine percent of those who were imaged, and early growth failure in seventy-four percent. The infants vomit, have diarrhea and malabsorption, develop massive hepatosplenomegaly and fail to thrive, which is why they are so often first labelled as cow's milk protein allergy. Ten infants received stem cell transplantation, and their median survival was only eight point six months, compared with three months without treatment.

Enzyme replacement with sebelipase alfa changed this. In the VITAL trial, six of nine infants, sixty-seven percent, were alive at twelve months, compared with none of twenty-one historical controls. Dosing began at zero point three five milligrams per kilogram weekly and was escalated up to five milligrams per kilogram when the response was poor. In the pooled final analysis of VITAL and a second study, reported by Vijay and colleagues, seventy-nine percent of nineteen infants survived to twelve months and sixty-eight percent to five years. In older children and adults, the ARISE trial by Burton and colleagues randomized sixty-six patients aged four and above. Alanine aminotransferase normalized in thirty-one percent on sebelipase alfa compared with seven percent on placebo. Fifty-eight percent had a low-density lipoprotein cholesterol of one

hundred ninety milligrams per deciliter or more at baseline, and ten of the thirty-two who had a biopsy already had cirrhosis.

The diagnosis is simple and cheap. Hamilton and colleagues described a dried blood spot assay that uses a specific inhibitor of the enzyme. Controls had activity between zero point five and two point three nanomoles per punch per hour, carriers between zero point one five and zero point four, and affected patients below zero point zero three. Send it for any infant with vomiting, diarrhea, hepatosplenomegaly and adrenal calcification, for any child with an enlarged liver, raised transaminases and an atherogenic lipid profile, and for any lean child labelled as having fatty liver. The 2017 guideline lists Wolman disease among the causes of neonatal liver failure and of hepatomegaly with a bright, echogenic liver, and LIPA belongs on every infant cholestasis panel.

Part eight. The mimics.

Several other diseases infiltrate or fill the infant liver. Acid sphingomyelinase deficiency causes a very large spleen and interstitial lung disease, and its biomarkers overlap with Niemann-Pick type C. For its non-neurological features there is now enzyme replacement with olipudase alfa, approved in 2022; in the adult trial spleen volume fell by about thirty-nine percent compared with essentially no change on placebo, and in eight children it fell by about forty-seven percent. Perinatal-lethal Gaucher disease presents above all with non-immune hydrops, sometimes with thickened skin and joint contractures. In a review of Gaucher disease presenting as neonatal cholestasis, seven of nine reported infants died.

The transient abnormal myelopoiesis of Down syndrome is a clonal proliferation of immature blood cells driven by mutations in the gene GATA1. In a European series of one hundred forty-six infants reported by Klusmann and colleagues, five-year event-free survival was sixty-three percent; early death was associated with a high white count, ascites, prematurity, bleeding and failure of spontaneous remission; and low-dose cytarabine helped high-risk infants. About one in five affected infants dies early, and liver fibrosis and liver failure carry the highest risk. A newborn with Down syndrome, blasts on the film, ascites and rising bilirubin has this disease until proven otherwise.

Stage 4S neuroblastoma can fill the liver of a young infant so massively that the child dies of respiratory or kidney failure. In a Dutch series of one hundred nineteen cases there were thirty-three deaths, twelve of them from massive hepatomegaly, almost all in infants diagnosed under four weeks of age, and mortality was sixty-nine percent when the MYCN oncogene was amplified beyond three copies, compared with twelve percent otherwise. Langerhans cell histiocytosis involves the liver in fifteen to sixty percent of children in different series; the liver is classed as a risk organ, mortality is thirty to fifty percent when it is involved, and sclerosing cholangitis can progress to cirrhosis and transplantation. Congenital leukemia, about two thirds of it myeloid, presents with skin nodules and hepatosplenomegaly.

Part nine. GALD and the meaning of a high ferritin.

GALD deserves a word in this chapter because it was long thought of as an iron storage disease. We now understand it as an alloimmune injury: maternal antibodies cross the placenta and attack the fetal liver, activating complement, and the iron overload is a consequence of fetal liver failure rather than its cause. The infant is in liver failure from birth, with severe coagulopathy, low albumin, high alpha-fetoprotein and relatively low transaminases. A ferritin above eight hundred micrograms per liter is sensitive but not specific, and a ferritin above twenty thousand should make you think of infection or HLH. The diagnosis is established by showing iron outside the liver, on magnetic resonance imaging of the pancreas or on a buccal mucosal biopsy. Staining the liver for the complement membrane attack complex, called C5b-9, supports the diagnosis but is not required.

Treatment works. Rand and colleagues reported that exchange transfusion combined with intravenous immunoglobulin gave about seventy-five percent survival without transplantation, compared with seventeen percent with the older antioxidant and iron chelation cocktail. And prevention works even better. The recurrence risk in later pregnancies is high, quoted between sixty-seven and ninety-two percent. In the series summarized by the National Health Service in England, among one hundred eighty-eight pregnancies treated with weekly intravenous immunoglobulin from about fourteen to eighteen weeks of gestation, five percent produced an affected liveborn infant, compared with forty-five percent of untreated pregnancies, and fetal loss fell from twenty-five percent to one percent.

Part ten. A practical approach.

Put this together as a sequence that runs alongside the standard cholestasis algorithm. On the first day, at the bedside, measure the spleen, look for fever, rash, skin nodules, rapid breathing, ascites and the features of Down syndrome, and look at a blood film. On the same day, send a blood count, ferritin, triglycerides, fibrinogen, soluble CD25, the international normalized ratio and alpha-fetoprotein, along with herpes and enterovirus polymerase chain reaction tests, and start acyclovir if the infant is ill. On the first or second day, send the storage tests: dried blood spot activities for lysosomal acid lipase and acid sphingomyelinase, the Niemann-Pick type C bile acid or lyso-sphingomyelin 509, and urine catecholamines. On the second day, use ultrasound to look for adrenal calcification and masses, and magnetic resonance imaging for iron outside the liver if the infant is in liver failure. Within the first week, send a rapid gene panel or genome that covers the HLH genes, NPC1 and NPC2, LIPA, SMPD1 and the Gaucher gene GBA1, and examine the marrow if the child is cytopenic.

Two principles run through this approach. First, do not wait for the liver biopsy. Most of these diagnoses are made on blood, and a biopsy in a coagulopathic infant adds risk; if one is done, ask for stains that identify Langerhans cells and macrophages, iron stains and attention to lipid storage. Second, sequence your actions rather than waiting for certainty: treat herpes empirically, start HLH therapy when the criteria are met, and send the enzyme activities before any transfusion that might confound them.

Part eleven. What the guidelines say.

How do the documents compare? On splenomegaly as a red flag, North America and Europe are aligned, because they share the 2017 joint guideline. On HLH there is a gap: the 2017 guideline does not name HLH at all, and the only diagnostic framework is the HLH-2004 criteria of the international Histiocyte Society. On Niemann-Pick type C biomarkers the documents differ: the joint guideline lists NPC1 in its table of causes but gives no advice on biomarkers, whereas the 2018 international consensus explicitly prefers the bile acid marker in neonatal cholestasis. On drug therapy for Niemann-Pick type C they also differ: arimoclomol and levacetylleucine were approved in the United States in 2024, while the European consensus predates them and relies on miglustat. On LAL deficiency they are aligned: LIPA is in the guideline table and sebelipase alfa is approved on both continents. Neither the American Association for the Study of Liver Diseases nor the European Association for the Study of the Liver has guidance on these infant disorders.

The material gap concerns GALD. The neonatal guideline mentions it only under repeated fetal loss, and it does not tell the hepatologist to arrange antenatal immunoglobulin for the mother's next pregnancy, even though the recurrence risk is around two in three or higher and the treatment largely prevents it. That referral is ours to make.

Part twelve. Pearls, traps and what to do differently.

Some traps recur. The first is calling a ferritin of ten thousand GALD in an infant whose transaminases are in the thousands; check soluble CD25, fibrinogen and herpes. The second is dismissing Niemann-Pick type C because the jaundice resolved; the cholestasis clears by three to four months, the spleen stays big, and the neurology comes years later. The third is trusting oxysterols in a cholestatic infant; they can be spuriously high, and they are also raised in acid sphingomyelinase deficiency and Wolman disease, so use the bile acid marker and genes. The fourth is listing an infant with active HLH for liver transplantation; a new liver does not fix the immune defect, so control the disease first and plan stem cell transplantation. And the fifth is calling vomiting, diarrhea and hepatomegaly in a six-week-old cow's milk allergy without an abdominal ultrasound, when calcified adrenals are the free clue to Wolman disease.

The numbers worth carrying are these. Ferritin above ten thousand is ninety percent sensitive and ninety-six percent specific for HLH. Thirty-eight percent of infants under two with acute liver failure at King's College had an HLH phenotype. Five-year survival in HLH is about sixty percent, and familial HLH needs stem cell transplantation. In Niemann-Pick type C, neonatal jaundice clears by three to four months and eight to nine percent die early. Untreated Wolman disease kills at a median of three point seven months, and sebelipase alfa brings first-year survival to between two thirds and four fifths. And in GALD, exchange transfusion with immunoglobulin gives about seventy-five percent transplant-free survival, while antenatal immunoglobulin cuts the rate of affected infants from forty-five percent to five percent.

So what should you do differently in clinic tomorrow? Palpate the spleen of every cholestatic infant and write down its size. When it is enlarged, add a storage and hematology arm to the work-up on the same day: ferritin, triglycerides, fibrinogen and soluble CD25; dried blood spot enzyme activities for lysosomal acid lipase and acid sphingomyelinase; the Niemann-Pick type C

bile acid marker; and a gene panel that includes the HLH, lysosomal and lipid genes. Treat herpes before you know. Order an abdominal ultrasound with a specific request to look at the adrenal glands. Read a ferritin together with the transaminases rather than on its own. And when a baby has had GALD, make sure the mother's obstetrician knows that her next pregnancy can be protected.